Sickle Cell Crisis

UP-20 | Universal Patient Care

Quick flow

01 Confirm Sickle Cell Disease + Assess [EMT / AEMT / Paramedic]

Usual crisis pattern • pain • chest symptoms • fever • neurologic findings • perfusion • pregnancy • priapism

02 Life-Threatening Complication? [EMT / AEMT / Paramedic]

Acute chest syndrome • stroke • sepsis • shock/severe anemia • splenic sequestration • priapism ≥4 hours

03 Support Oxygenation + Temperature [EMT / AEMT / Paramedic]

Oxygen for hypoxemia or respiratory distress • target SpO_2 ≥95% unless documented lower baseline • keep warm

04 Treat Pain Promptly [EMT / AEMT / Paramedic]

Nonpharmacologic care • adult UP-11 • approved pediatric weight-based pathway • reassess pain and sedation

05 Fluids Only When Indicated [AEMT / Paramedic]

No routine bolus • hypotension/poor perfusion: adult 250 mL or pediatric 10 mL/kg isotonic aliquots with reassessment

06 Rapid Transport When High Risk [EMT / AEMT / Paramedic]

Chest symptoms • hypoxemia • neurologic deficit • fever ≥38.5°C • shock/anemia • pregnancy • atypical or uncontrolled pain

Overview

- Use this protocol for a patient with known or suspected sickle cell disease and acute pain or illness. Sickle cell trait alone does not establish a vaso-occlusive crisis.
- Do not assume every complaint is an uncomplicated pain crisis. Screen for acute chest syndrome, stroke, fever or sepsis, shock or severe anemia, splenic sequestration, priapism, pregnancy complications, dehydration, and pain that differs from the patient's usual crisis.
- Treat pain promptly and respectfully while preserving airway, ventilation, perfusion, neurologic assessment, and rapid access to definitive care.

Provider-Level Actions

Recognize complications, begin supportive care, treat pain early, and avoid unnecessary delay or excess fluid.

EMT

- Perform XABCDE, obtain two complete sets of vital signs when feasible, assess pain, temperature, SpO_2, mental status, perfusion, and the patient's usual crisis pattern and prior complications.
- Ask specifically about chest pain, cough, fever, dyspnea, new weakness or speech change, syncope, pregnancy, abdominal fullness, and priapism duration when applicable.
- Keep the patient warm, position for comfort, provide oxygen for hypoxemia or respiratory distress, and support ventilation when needed.
- Use approved nonpharmacologic and EMT pain measures. Do not dismiss or delay care because the patient appears calm or has frequent EMS encounters.

AEMT

- Perform all EMT care. Establish IV access when indicated without delaying transport or analgesia.

- Do not give a routine crystalloid bolus for uncomplicated vaso-occlusive pain. For hypotension or poor perfusion, administer isotonic crystalloid in reassessed aliquots: adult 250 mL or pediatric 10 mL/kg.
- For suspected dehydration without hypotension, use only KVO or maintenance-rate fluid during transport and reassess for pulmonary congestion.
- Administer approved AEMT analgesia under the applicable adult or pediatric pain pathway and monitor treatment response.

Paramedic

- Perform all prior care. Treat pain promptly: use UP-11 for age 16 or older and PM-04 for age 2 through younger than 16. For a child younger than 2, provide nonpharmacologic care, obtain Medical Control direction for analgesia within scope, and do not use an adult-dose fallback. PM-04 authorizes no additional AEMT standing-order analgesic beyond its EMT measures. Follow PM-04 medication, route, dose, contraindication, monitoring, and repeat-dose limits.
- For opioid or ketamine analgesia, continuously monitor airway, respiratory effort, SpO_2, cardiac rhythm, and waveform $EtCO_2$ when feasible; keep suction and BVM immediately available.
- Obtain a 12-lead ECG for adult chest pain, dyspnea, syncope, abnormal rhythm, or another cardiac concern without delaying transport.
- Provide early receiving-facility notification for acute chest syndrome, stroke findings, fever, shock, suspected severe anemia or sequestration, prolonged priapism, pregnancy concern, or an atypical crisis.

High-Risk Complications

A normal appearance or familiar pain complaint does not exclude a time-critical complication.

EMT

- Acute chest syndrome: suspect with chest pain, cough, fever, dyspnea, abnormal lung sounds, or new hypoxemia. Provide oxygen and ventilation support as indicated and transport rapidly.
- Stroke: treat any new focal deficit, speech change, severe disequilibrium, vision change, or altered mental status under the suspected-stroke pathway.
- Fever: temperature 38.5°C / 101.3°F or higher is high risk for invasive infection and requires prompt emergency-department evaluation.
- Splenic sequestration or severe anemia: suspect with weakness, pallor, tachycardia, abdominal fullness or splenic enlargement, syncope, hypotension, or poor perfusion.
- Priapism: duration four hours or longer is a time-critical emergency; severe pain or a rapidly progressing episode also requires urgent evaluation.

AEMT

- For hypotension or poor perfusion, provide small reassessed isotonic fluid aliquots and stop or slow fluid if respiratory distress, crackles, or worsening oxygenation develops.

Paramedic

- Coordinate rapid specialty-capable transport and early notification. Provide cause-specific airway, shock, seizure, stroke, or cardiac care under the applicable Claiborne protocol.

Indications

- Known or suspected sickle cell disease with acute pain, fever, respiratory symptoms, neurologic symptoms, weakness, syncope, priapism, pregnancy concern, or other acute illness.
- Concern for a sickle cell complication even when the presenting pain resembles a prior vaso-occlusive crisis.

Contraindications

- Do not diagnose vaso-occlusive crisis solely from sickle cell trait.
- Do not give routine crystalloid boluses to an otherwise perfusing patient with uncomplicated pain.

Assessment | Focused history

- Confirm sickle cell disease type when known, prior acute chest syndrome or stroke, baseline SpO_2, transfusion history, usual pain location and severity, and the patient's previously effective and tolerated treatment.
- Determine onset, location, severity, and character of pain and whether this episode differs from the patient's usual crisis.
- Ask about fever, chills, cough, chest pain, dyspnea, focal neurologic symptoms, syncope, reduced intake, vomiting or diarrhea, pregnancy, abdominal swelling, and priapism duration.
- Review allergies, analgesics already taken, hydroxyurea or other disease-modifying therapy, anticoagulants, recent hospitalization, and indwelling vascular access.

Assessment | Focused examination and monitoring

- Obtain two complete sets of vital signs when feasible, including temperature, SpO_2, mental status, pain score, respiratory effort, and perfusion.
- Assess lung sounds, work of breathing, skin color and temperature, hydration, pulse quality, neurologic status, and the painful area for an alternate cause.
- Examine the abdomen for tenderness or enlargement when severe anemia or splenic sequestration is suspected; do not delay transport for a prolonged examination.
- Use continuous pulse oximetry for chest or respiratory symptoms. Add cardiac monitoring, 12-lead ECG, and waveform $EtCO_2$ when indicated by the presentation or treatment.

Treatment steps

- Treat pain promptly: use UP-11 for age 16 or older and PM-04 for age 2 through younger than 16. For a child younger than 2, provide nonpharmacologic care, obtain Medical Control direction for analgesia within scope, and do not use an adult-dose fallback. PM-04 authorizes no additional AEMT standing-order analgesic beyond its EMT measures. Follow PM-04 medication, route, dose, contraindication, monitoring, and repeat-dose limits.
- Perform Universal Patient Care and immediately address airway, ventilation, hypoxemia, shock, altered mental status, seizure, stroke findings, or another life threat.
- Provide oxygen for hypoxemia or respiratory distress and target SpO_2 at least 95% unless the patient has a documented lower baseline. Keep the patient warm and avoid cold exposure.
- Treat pain promptly. Use UP-11 for patients 16 years or older and the approved pediatric age- and weight-based pathway for patients younger than 16 years.
- Do not withhold analgesia because the patient appears calm, has recurrent crises, or has received opioid medication previously. Use the patient's known effective regimen only when it is consistent with the approved protocol, formulary, and current clinical condition.
- Do not give a routine fluid bolus for uncomplicated pain. For hypotension or poor perfusion, give isotonic crystalloid in reassessed aliquots: adult 250 mL or pediatric 10 mL/kg. For dehydration without hypotension, use only KVO or maintenance-rate fluid during transport.
- For suspected acute chest syndrome, support oxygenation and ventilation, apply continuous SpO_2 and cardiac monitoring, obtain a 12-lead ECG when indicated, notify early, and transport rapidly without excessive fluid.
- For temperature 38.5°C / 101.3°F or higher, provide supportive sepsis care and prompt transport. No field antibiotic is authorized under this protocol.
- Use the suspected-stroke pathway for any new focal neurologic deficit. Transport urgently for suspected severe anemia or splenic sequestration and for priapism lasting four hours or longer.
- Strongly recommend transport for acute chest symptoms, hypoxemia, neurologic deficit, fever, hypotension, poor perfusion, suspected severe anemia or sequestration, prolonged priapism, pregnancy concern, uncontrolled or atypical pain, or any opioid or ketamine administration.

Warnings / precautions

- Do not allow a familiar pain presentation to obscure acute chest syndrome, stroke, sepsis, shock, severe anemia, splenic sequestration, or another emergency.
- Avoid excessive crystalloid; overhydration can worsen pulmonary complications.
- Opioid or ketamine analgesia requires continuous airway, respiratory, oxygenation, cardiac, and $EtCO_2$ monitoring as specified by the applicable pain protocol.
- Do not use stigmatizing language such as 'drug-seeking' in the clinical record. Document objective findings, reported pain, treatment, response, and safety concerns.

Clinical pearls

- Acute chest syndrome may initially resemble pneumonia or an uncomplicated pain crisis and can progress rapidly.
- Pain behavior varies widely. A calm appearance does not reliably measure pain severity.
- A patient's prior effective treatment is clinically useful, but medication selection and dose must remain within the approved Claiborne protocol and formulary.
- Priapism lasting four hours or longer is an emergency because delay increases the risk of permanent dysfunction.

Special populations | Pediatric patients

- For Claiborne clinical protocols, pediatric medication dosing applies to patients younger than 16 years. Use actual or length-based weight.
- Children may deteriorate rapidly from infection, acute chest syndrome, aplastic crisis, or splenic sequestration. Fever 38.5°C / 101.3°F or higher requires prompt emergency evaluation.

Special populations | Pregnancy

- Use left lateral positioning when appropriate, monitor closely for hypoxemia and shock, and coordinate obstetric-capable destination and early notification.

Related pathways / contacts

Open UP-11 Adult Pain Control - Use the approved adult analgesia pathway for patients 16 years or older.

Source and references

American Society of Hematology - Sickle Cell Disease: Management of Acute Complications pocket guide. | American Society of Hematology 2020 Guidelines for Sickle Cell Disease: Management of Acute and Chronic Pain; guideline reviewed in 2023. | National Heart, Lung, and Blood Institute - Sickle Cell Disease symptoms and emergency complications. | Tennessee EMS Protocol Guidelines, September 2025 edition. | Claiborne EMS UP-11 Pain Control - Adult and approved pediatric medication formulary.

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